

Rojin Roy

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Professional Summary

Embedded Systems Engineer with 4+ years of experience developing embedded C firmware for PIC32-based motion-control systems. Experienced in hardware-software integration, communication protocols (UART, SPI, TCP/IP), and system-level debugging using oscilloscopes, logic analyzers, and firmware logs. Hands-on experience with embedded Linux, Linux kernel builds, U-Boot, and BeagleBone Black bring-up.

Professional Experience

Unistring Tech Solutions Pvt. Ltd.

Application Engineer

Nov 2025 – Present

Junior Application Engineer

Mar 2022 – Nov 2025

- Developed embedded C firmware for PIC32-based motion-control systems.
- Implemented UART communication features and diagnostics handling.
- Debugged Ethernet (TCP/IP) and UART communication using firmware logs and hardware tools.
- Developed C-based production test applications using Win32 APIs.
- Performed hardware-level debugging using oscilloscope and logic analyzer.
- Integrated motors, sensors, and peripheral interfaces in embedded systems.
- Supported firmware validation, testing, and regression cycles.

Projects

Linux Kernel Build and Bring-up (BeagleBone Black)

- Built U-Boot and the Linux kernel for BeagleBone Black using an ARM cross-compilation toolchain.
- Booted the target using the custom-built bootloader and kernel, and verified system startup through the serial console.
- Analyzed kernel boot logs (`dmesg`) to verify hardware initialization and boot sequence.

BeagleBone Black PRU Bring-up using remoteproc

- Configured and built a Linux kernel with `remoteproc` support for the PRU subsystem.
- Brought up the PRU subsystem on BeagleBone Black and verified firmware loading through the `remoteproc` framework.
- Debugged subsystem initialization using kernel logs and the Linux `sysfs` interface.

Technical Skills

Languages: C

Platforms: PIC32, Linux

Protocols: TCP/IP, Ethernet, UART, SPI, I²C

Concepts: Interrupts, Timers, PWM, State Machines

Tools: Git, Linux, Oscilloscope, Logic Analyzer

Domain Knowledge: Motion Control, Embedded Systems, System Integration, Firmware Validation

Education

Bachelor of Technology (B.Tech)

Guru Nanak Institute of Technology, Hyderabad

2017